XIAOTIAN CHEN

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EDUCATION

Columbia University New York, NY
M.S. in Electrical Engineering Expected Dec 2027

The University of Nottingham Ningbo China Ningbo, China
Bachelor of Engineering in Electrical and Electronic Engineering Graduation Date: Jul 2026
GPA: 3.5/4.0
Classification: Upper Second-Class Honours (2:1).

WORK EXPERIENCE

Ningbo Ausstattung Machinery Technology Co., Ltd.- , R&D Department Ningbo, China
Algorithm Engineer Intern Jun 2026 - Aug 2026

- Created a Python-based control application for five independently addressable joint motors in a articulated inspection robot, combining real-time position, velocity, and torque feedback over CAN/CAN-FD.
- Implemented motor enable/disable, emergency stop, software limits, per-motor speed profiles, zeroing, and asynchronous communication handling to support safe multi-joint operation.
- Established a 20 Hz motion-recording and playback pipeline operating timestamp JSON trajectories, interpolation, keyframe editing, smooth pose transitions, and 0.5×-2× playback speeds.
- Designed full-system integration and testing of cable-driven multi-joint mechanisms: analyzed friction, backlash, load variation, mechanical interference, and firmware/communication compatibility issues.

RESEARCH

Multi-Static Passive Wi-Fi Human Activity Recognition with CSI Aug 2025 - Jun 2026
Independent Researcher

- Constructed a multi-static passive Wi-Fi sensing platform using three Raspberry Pi 4B nodes, Nexmon CSI, and TCP- based synchronization for coordinated CSI data acquisition.
- Established an end-to-end CSI preprocessing and dataset pipeline with packet filtering, subcarrier selection, windowing, and sample labeling to improve signal quality and consistency.
- Trained PyTorch-based human activity recognition models and evaluated single-, dual-, and three-node sensor fusion strategies across multi-node configurations.
- Investigated performance degradation in multi-node fusion and developed a weighted decision-level fusion strategy: analyzed cross-scenario domain shift to improve model generalization.

Advanced Modulation and Control for Multi-Active-Bridge (MAB) Converter Jun 2025 - Aug 2025
Researcher

- Researched and accomplished advanced strategies for MAB converters, including dead-time optimization, DC-AC modulation, decoupled control algorithms, and equivalent modeling of multi-winding transformers.
- Built converter and system-level simulations in PLECS, MATLAB/Simulink, and LTspice; analyzed power-transfer efficiency, device stress, voltage/current waveforms, and closed-loop stability to guide control design.
- Assisted to hardware design and bring-up: schematic review, PCB component soldering, oscilloscope/logic-analyzer debugging; integrated signal-acquisition circuitry, control board, and main power stage into a working prototype.

Indoor Inspection Robot-Autonomous Navigation System Jul 2024 - Sep 2024
Researcher

- Designed and deployed a ROS 2-based autonomous navigation stack; fused LiDAR, IMU, and wheel-odometry data and implemented SLAM, reducing localization error from 0.30 m to 0.05 m.
- Developed obstacle detection and avoidance using 10 Hz LiDAR scans, achieving reaction time under 0.5 s and sustaining 4 hours of collision-free indoor operation.
- Authored a Python status-monitoring node tracking sensor health and battery state; achieved ~95% alert precision during extended runs.

Development of UAV Autonomous Target Search and Navigation System May 2024 - Jul 2024
Researcher Assistant

- Built Ubuntu 22.04+ROS 2 Humble+PX4 dev environment, developed ROS 2 nodes for TurtleBot/Gazebo, integrated pub-sub functions to optimize structure, cutting resource use & ensuring communication compatibility.
- Iterated path planning & controllers (A* for square trajectories + PID optimization, RRT*+MPC for circular flights), solving trajectory deviation and smoothness issues, achieving ±0.2m (square)/±0.1m (circular) error, speed fluctuation < 5%.
- Integrated visual SLAM, YOLOv8, multi-sensor+EKF fusion, built UAV autonomous target search & navigation system, enabling autonomous flight & high-precision trajectory tracking in simulation/real environments.

PROJECTS

Design and Build a Switch Mode Power Supply (SMPS) Sep 2023 - May 2024
Project Leader

- Calculated optimal configuration of transformers, inductors, capacitors, and resistive loads, ensuring minimal voltage ripple and stable output.
- Designed a closed-loop forward converter on KiCad, implementing with MOSFET drivers enabled by PWM.
- Constructed a stable closed-loop forward converter model with MOSFET drivers and implemented PWM for efficient power regulation; maintained a stable 10 V DC output from a 35 V unregulated AC power supply through design and debugging, ensuring performance under varying loads.

Doppler Radar System Design and Implementation Sep 2023 - May 2024
Project Leader

- Designed a bandpass filter for radar frequency extraction; prototyped on breadboard, validated with signal generator and oscilloscope, and finalized by soldering.
- Programmed STM32L476RG MCU with STM32 Cube IDE to perform Fast Fourier Transform (FFT) for signal analysis.
- Constructed LCD based on CPLD in Xilinx ISE, established communications between LCD and STM32 through RS- 485 and UART, and coded decimal-to-binary conversion for CPLD input signals.

Intelligent Robot Design Competition of UNNC Oct 2023 - Jan 2024
Project Leader

- Led an interdisciplinary team to devise an autonomous robot, oversaw assembly and Arduino programming, and won 1st prize.
- Introduced PID closed-loop control, used Hall encoders for real-time wheel speed and dynamically adjusted motor PWM, eliminating traditional delay-induced trajectory drift and boosting symmetry to over 98%.
- Tuned PID via Ziegler-Nichols method and optimized four-wheel speed ratio/delay to adapt to friction/voltage changes, accomplishing 5-minute continuous "8"-shaped travel (deviation <±2cm, speed fluctuation <±3%).

LANGUAGE AND IT SKILLS

Languages: Python, C++, C, MATLAB.
Embedded/Robotics: ROS 2, PX4, STM32, Arduino, CAN/CAN-FD, UART, I2C.
Tools: Linux, KiCad, PLECS, LTspice, MATLAB/Simulink, PyTorch, SolidWorks.